



Dependence, novelty, autonomy and the emergence of spacetime

Aiden Meyer¹

Received: 12 May 2025 / Accepted: 4 May 2026
© The Author(s) 2026

Abstract

Emergence is often defined in terms of dependence, autonomy, and/or novelty. However, the debate surrounding the emergence of spacetime in quantum gravity has typically proceeded without much clarity regarding the definition of emergence and how dependence, novelty, and autonomy are to be understood in this context. To shed light on this, I introduce some desiderata for emergence criteria in quantum gravity, namely (i) avoiding spatiotemporal presuppositions, (ii) accommodating both flat and hierarchical emergence cases, (iii) clarity and precision and (iv) metaphysical neutrality. Then I compare three main approaches to defining dependence, novelty, and/or autonomy, two of which are present in the existing spacetime emergence literature. These are, firstly, various approaches that define emergence in terms of the elimination of degrees of freedom (Bain 2013a, 2013b, Wilson 2020, Crowther 2018), and secondly, Crowther's, (2021, 2022) view. Then, a third approach, which defines emergence primarily in terms of conditional probability (Franklin 2024, Franklin and Robertson 2024), that has yet to be applied to the spacetime emergence problem, is introduced and applied to key quantum gravity cases. I argue that, unlike the other views, this third option fulfils all the desiderata and is the most generally applicable of the options discussed here.

Keywords Spacetime emergence · Quantum gravity · Philosophy of physics · Emergence · Philosophy of science

✉ Aiden Meyer
aiden.meyer@student.unimelb.edu.au

¹ School of Historical and Philosophical Studies, The University of Melbourne, Victoria, Australia

1 Introduction

Much research in quantum gravity (QG) indicates that spacetime may be non-fundamental.¹ As such, it is often claimed that spacetime is emergent in at least some QG theories (Huggett and Wüthrich, 2013; Lam et al., 2018; Crowther, 2016, 2021, 2022). Even if we accept the possibility of non-fundamental spacetime in QG, the status of this emergence claim is debated, as an eliminativist option regarding spacetime in QG has been both suggested (Le Bihan, 2018) and endorsed (Baron, 2020; Baron & Le Bihan, 2021).² To make progress on the spacetime emergence issue, it is useful to have clear criteria for what it takes for spacetime to emerge. Yet, so far, this debate has typically proceeded without much clarity regarding the definition of emergence itself.³ Having clearly defined emergence criteria would also help assess the status of emergence in any new proposals for QG theories.

There is a general understanding of emergence, namely, that it involves something (a theory, entity or property) that displays dependence on its base, yet also displays autonomy (or independence) and/or novelty with respect to this base (Humphreys, 2016; Crowther, 2016, 2021; Baron, 2020; Wilson, 2020). However, there is little consensus regarding how each of these key features of emergence are to be understood in the general (non-QG-specific) literature (Baron, 2020). Crucially, we need precise statements of these notions specifically optimised for the spacetime case, and it is not apparent that we can just carry these over from other cases, as many extant accounts of emergence presuppose spatiotemporal assumptions inappropriate in a QG context.

Two main options for defining emergence (in the sense of dependence novelty and autonomy) in a QG-specific context have been offered: firstly, Crowther's (2021, 2022) view (a refinement of an earlier position expressed in Crowther (2016)) and secondly, various approaches defining emergence in terms of elimination of degrees of freedom (DOFs) (endorsed by Bain (2013a, 2013b) and Wilson (2020), the latter of which is applied to QG by Crowther (2018)). I offer a third option, yet to be applied to QG. This builds on, but generalises, prior work by Franklin (2024) and Franklin and Robertson (2024) and defines dependence and autonomy via *conditional probability* and novelty in terms of *formal or dynamical distinctness*. I call this approach the *conditional probability-based approach*.⁴

¹ See Huggett and Wüthrich (2013) and Lam et al. (2018) for claims of spacetime's non-fundamentality in loop quantum gravity and causal set theory. Huggett (2017), Vistarini (2019), and Huggett and Wüthrich (2025) argue for spacetime's non-fundamentality in string theory, and Crowther (2016) for a variety of other QG approaches.

² See also Ismael (2021).

³ Crowther (2016, 2018, 2021, 2022) work is a key exception, as is Bain (2013a).

⁴ Here, I remain neutral regarding whether emergence is occurring between theories (theoretical emergence) or things in the world (ontological/metaphysical emergence). While my discussion focuses on theories and the posits of theories (theoretical objects, properties and so on), those of a realist persuasion regarding non-fundamental ontology could take the dependence, autonomy and novelty criteria given here as criteria for inclusion in a realist ontology. Franklin and Robertson (2024) do so regarding their version of the conditional probability-based approach, as do Wilson's (2020) and Bain (2013a, 2013b) emergence views considered in Sect. 3–4. However, I will not explore this issue in detail here.

This article proceeds as follows. In Sect. 2, I first introduce the flat/hierarchical emergence distinction, then I provide four desiderata for generally applicable candidate definitions of dependence, novelty and autonomy in QG. These are (i) *avoiding spatiotemporal presuppositions*, (ii) *accommodating both flat and hierarchical emergence cases*, (iii) *clarity and precision* and (iv) *metaphysical neutrality*. In Sect. 3, I assess Bain (2013a, 2013b) and Wilson's (2020) views in light of these desiderata and find that neither fulfils (iv), the metaphysical neutrality criterion, and Bain's view also does not fulfil (ii), the accommodating both flat and hierarchical emergence desideratum. However, Bain's view of novelty is on the right track. Then Crowther's (2022) view is assessed in Sect. 4. This view fares better regarding these desiderata and is largely on the right track, but struggles to fulfil (iii), the clarity and precision desideratum. In Sect. 5, I introduce the conditional probability-based approach to dependence and autonomy, an approach yet to be applied to spacetime emergence in QG, and show that it fulfils all four desiderata, unlike the existing approaches considered in Sect. 3–4, and is a more generally applicable account, and thus one with greater general utility. The approach builds on the work of Franklin and Robertson (2024) and Franklin (2024). However, their extant conditional probability-based strategy requires modification to allow it to accommodate both *flat* (intra-level emergence) and *hierarchical* emergence (inter-level emergence) such that it can fulfil (ii), the flat and hierarchical emergence desideratum, as both types of cases may exist in QG (Crowther, 2021). Thus, a generalised conditional probability-based approach that can accommodate flat and hierarchical emergence is introduced.⁵ This is combined with the previously mentioned understanding of novelty as formal distinctness. I then apply these criteria to three QG cases: loop quantum gravity, string theory (notable hierarchical emergence cases), and quantum graphity (a flat emergence case).

2 Introducing spacetime emergence

2.1 Spacetime, QG and flat and hierarchical and emergence

To compare candidate emergence criteria sets, we need desiderata for emergence criteria in QG. An explicit list of such desiderata is absent in the existing literature. But for my proposed desideratum to be intelligible, there is one crucial distinction that I need to introduce first. This is the distinction between flat (with-level) and hierarchical (between-level) emergence in QG (Crowther, 2021).⁶ This distinction is best illustrated via examples.

A key example of hierarchical emergence, according to Crowther (2021), is loop quantum gravity (LQG). For Rovelli (2020), LQG is definable in terms of a Hilbert space (H), an operator algebra and a collection of transition amplitudes. This Hilbert space has a basis, the spin network basis, whose states are $|\Gamma, j_i, v_n\rangle$, where Γ is “an

⁵ Similar novelty-related views appear in Knox (2016), Franklin and Knox (2018), Franklin (2024), Crowther (2015, 2016) and Bain (2013a, 2013b).

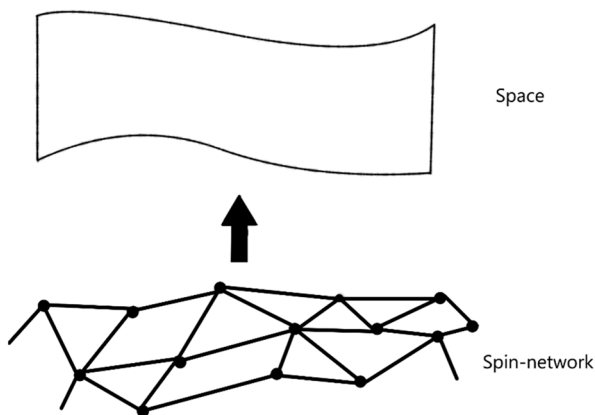
⁶ Levels are, following Crowther (2021, 2022) account of flat and hierarchical emergence, understood in terms of differing energy scales or length scales.

(abstract) combinatorial graph” to which we assign a “discrete quantum number j_l for each link l of the graph, and a discrete quantum number v_n for each node n of the graph” (Rovelli, 2020, p. 129).⁷ Rovelli (2020, p. 129) interprets the graph’s nodes as the basic “quanta of space”. The links determine the adjacency of these. Both the area of the surfaces that separate a pair of nodes and the volume of the quanta are discrete. The volume is determined by v_n , and the area by j_l (Rovelli, 2020). Lam and Wüthrich’s (2018) interpretation of LQG regards spin-networks as LQG’s basic posits.⁸

LQG states are typically superpositions of spin-network states. Additionally, two events may be adjacent in a spin-network structure but correspond to events in ordinary spacetime that are arbitrarily far apart, leading to a disordered locality. For Lam and Wüthrich, these features mean we should not view LQG’s fundamental ontology as fundamentally spatiotemporal. According to this interpretation, if spacetime exists, it emerges hierarchically at a higher level. Hierarchical emergence is the non-spatiotemporal version of synchronic emergence (Crowther, 2021).⁹

For an idea of how this hierarchical recovery of space in canonical LQG works (as depicted in Fig. 1), consider weave states. Weave states are (semi-classical) spin-network states $|S\rangle$ that are eigenstates of the area operator $\hat{A}(S)$ and volume operator

Fig. 1 Space hierarchically emerging from a lower-level spin-network structure (image adapted from Crowther (2021))



⁷To be more precise the links are colored by spins (group representations, usually of $SU(2)$ and the nodes with intertwiners, where we can think of the latter as a “special vector which intertwines the vector spaces associated with the colorings of the different links” (Norton, 2020, p. 17; Rovelli & Vidotto, 2014). As Livine (2024, p. 4) describes, “[i]ntertwiners at the nodes represent excitations of the 3d volume. And spins give the quanta of area of the interfaces between neighbouring chunks of volume”.

⁸Norton (2020) highlights that this isn’t LQG’s only possible interpretation and lists alternatives. Some retain spacetime, and others do not. Here, I discuss LQG accepting the common assumption that spacetime is non-fundamental in this theory. However, I am neutral regarding the general metaphysics of LQG (and the metaphysical interpretation of spin-networks). Additionally, as I just described LQG’s kinematics, namely “the spin network states” that “provide a basis for the kinematic Hilbert space ... rather than the dynamics,” we should, following Crowther (2021, p. 7924), speak of “space (rather than spacetime)” as “fundamentally constituted by a spin network”.

⁹Synchronic emergence is emergence occurring at a higher level resulting from lower-level components in a system, where this emergence happens synchronically (at the same time). Whereas diachronic emergence involves the generation of novel and/or autonomous features via the system’s temporal evolution, occurring where this transformation of the system occurs at the same level (Crowther, 2021). Crowther (2021) draws on Guay and Sartenauer (2016) for this account of diachronic and synchronic emergence.

$V(R)$ of LQG, with eigenvalues approximating classical spatial area and volume, as given by the classical 3-metric g . More precisely, if we “select a classical macroscopic 3d gravitational field e , which determines a macroscopic 3d metric” (Rovelli, 2004, p. 268), then at low energies, a spin network approximating this classical metric can be constructed (called a weave state), where this approximation holds at a scale ℓ (which corresponds to a higher level) greater than the (low-level, and high energy) Planck length scale ℓ_P . Let’s demand that $|S\rangle$ is an eigenstate of the volume and area operators of LQG ($V(R)$, and $A(S)$ respectively) with eigenvalues equal to the volume of a macroscopic 3-region R (and of the area of macroscopic 2-surface S) “determined by e , up to small corrections in ℓ_P/ℓ (Rovelli, 2004, p. 268). In more formal terms:

$$A(S) |S\rangle = (A[e, S] + O[\ell_P/\ell]) |S\rangle \quad (1)$$

$$V(R) |S\rangle = (V[e, R] + O[\ell_P/\ell]) |S\rangle \quad (2)$$

Here $V[e, R]$ is the volume of the region R determined by the gravitational field e , and $A[e, S]$ the area of the surface S , as determined by e . The basic idea encapsulated by the above equations is that at sufficiently low energies (corresponding to a different macroscopic, classical seeming level/energy scale), weave states numerically agree with the metric of General Relativity (GR), but diverge from it at high energies where the higher-order terms become more significant.¹⁰

We can contrast this with flat (within-level) emergence (Crowther, 2021, 2022), a potential case of which is quantum graphity. Here, spacetime (more precisely, geometry) emerges via a phase transition from a ‘prior’ non-spatiotemporal (or non-geometric) state, where both states exist on the same level.¹¹ Flat emergence is the non-spatiotemporal analogue of diachronic emergence: emergence involving a within-level transformation of the system (e.g. Humphreys, 2016), but without presupposing spacetime. We can represent the basic structure of quantum graphity as a graph with points representing events related by links between these, and the dynamics are understood in terms of changes in the connections between points (Crowther, 2022).¹²

[In Quantum graphity] ... [t]he micro-description of the early (pre-geometric) universe is understood as a complete graph ... this is a high-energy, maximally-connected state ... [where] ... the dynamics ... [are] ... invariant under permutation of the events, and, because the entire universe is one-link adjacent to any

¹⁰ This presentation of weave states is not diffeomorphism invariant, however, we can construct analogous weave states via diffeomorphism invariant s -knots (sometimes called abstract spin networks), where, “the s -knot state $|s\rangle = P_{diff} |S\rangle$ is called the weave state of the 3-geometry $[g]$, the equivalence class of 3 metrics to which the metric g belongs.” P_f (Rovelli, 2004, p. 269). Ultimately, we should be speaking of s -knots/abstract spin networks if treating these as physical entities/structures (Norton, 2020), an assumption implicit hereafter.

¹¹ Loop quantum cosmology, causal set cosmology, and other pregeometric approaches to QG are potential cases of flat emergence (Crowther, 2021, 2022).

¹² See Konopka et al. (2008) for further details.

event, there is no notion of geometry or locality. The micro-degrees of freedom are the states of the links, and these evolve in time under the Hamiltonian. As the universe cools and condenses, it undergoes a phase transition in which many of the connections switch off. The system at low-energy ... is a graph with far fewer edges ... the permutation invariance breaks, and instead translation invariance arises. At this stage locality is ... define[able], and we gain a sense of relational geometry[Thus] ... geometry emerges in this phase transition, known as geometrogenesis. (Crowther, 2021, p. 7300)

We should note that, unlike some other spacetime emergence cases, there is a fundamental notion of time in quantum graphity: a ‘before’ and ‘after’, but no fundamental spacetime geometry (Crowther, 2021).

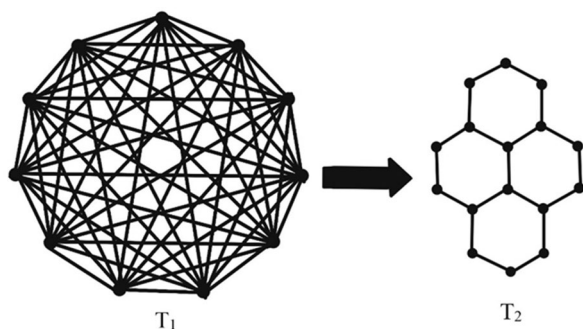
Based on the above example (and Fig. 2), we can see how spatiotemporal geometry can emerge from a prior non-geometric state in the within-level sense required of flat emergence.

2.2 Desiderata for generally applicable accounts of spacetime emergence in QG

With this basic introductory material out of the way, to identify the presence of, and clarify the definition of emergence cases like these, we need criteria for emergence. However, to compare these definitions of emergence, we need some desiderata. However, I don’t take these to indicate that we shouldn’t use accounts that do not fully fulfil these desiderata. These accounts, which I argue do not fully fulfil the desiderata (e.g., Bain’s, Wilson’s, and Crowther’s), are interesting and can still be used in the context in which they best apply. But, if one is interested in coming up with a broadly applicable account of emergence that is useful for interpreting spacetime emergence in multiple cases, then these desiderata are useful.¹³

Having a widely applicable approach to defining emergence is beneficial for a few reasons. Firstly, there are practical benefits. If we can give a good account of emergence with a single metaphysical tool, this is easier than drawing on many different philosophical tools when considering emergence in QG. Focusing on one generally applicable definition provides the philosophical community with a clearer picture

Fig. 2 The flat emergence of geometry at the same level. Here, T_1 is the high-energy maximally-connected pre-geometrogenesis state, T_2 is the geometric low-energy state, with fewer connections resulting in emergent locality (image adapted from Crowther (2021), and Konopka et al. (2008))



¹³ I thank an anonymous reviewer for suggesting that I clarify the role of the desiderata here.

of what we mean when we use the term emergence in QG. This reduces the risk of misunderstanding due to the implicit use of different definitions of emergence, as well as simplifies discussions of emergence and reduces the need to do the work of introducing customised definitions for each case. Having a shared, generally applicable definition of emergence is also useful when comparing instances of emergence across different QG cases, as it helps identify similarities and differences between these cases.¹⁴

Secondly, to the extent that one is concerned with parsimony regarding the number of conceptual tools used in philosophy of science, having a generally applicable account is beneficial in this regard. Basically, if we can define emergence across multiple QG cases with a single definition, there are parsimony-related benefits to focusing on that definition (or at least having such a definition available), rather than using a wider variety of definitions.

While this paper uses the term ‘spacetime emergence’ because it is a common framing in the existing literature, my approach is not committed to the existence of a single unified ‘general problem of spacetime emergence’, a notion critiqued by Jaksland and Salimkhani (2023). It is only committed to the idea that having a conceptual tool (namely, a definition of emergence, i.e., dependence, novelty and autonomy) that is useful in multiple QG cases where emergence of some sort may occur is beneficial.¹⁵ Indeed, the results of this paper are easily adapted to Jaksland and Salimkhani’s approach of focusing on the emergence of specific spatiotemporal aspects. This is because having a definition of emergence (i.e., dependence, novelty, and autonomy) that can be applied to understanding the emergence of specific spatiotemporal aspects in different QG cases still seems like a useful thing to have for the reasons mentioned previously, regarding simplicity/ease of use and conceptual parsimony.¹⁶ So, those who favour Jaksland and Salimkhani’s (2023) approach to discussing emergence in QG can also benefit from the findings of this article.

With this said, I propose the following desiderata for generally applicable criteria for spacetime emergence in QG:

- (i) These criteria should not presuppose spatial, temporal or spatiotemporal notions, either explicitly or implicitly.

¹⁴This emphasis on practical inferential and heuristic benefits as a reason to prefer particular conceptual tools (such as emergence definitions) has affinities with Wallace (2022, 2024a, 2024b) view. Wallace understands conceptual philosophical tools as providing second-order, heuristically and inferentially beneficial and perspicuous pictures of what mathematised physical theories tell us about reality. However, I am not necessarily committed to all the specifics of Wallace’s methodological account in this paper (although I personally have a favourable attitude toward Wallace’s approach as a whole (e.g., Legg and Meyer (2026))).

¹⁵Jaksland and Salimkhani (2023, p. 17) could be interpreted as suggesting that the question of “what emergence is in general” is a distinct question from arguing for the idea of the ‘general problem of spacetime emergence’. I am interested in the former question rather than the latter, so I don’t think that our approaches are incompatible.

¹⁶We could, for example, even think of my discussion of quantum graphity (where both the conditional probability-based approach and Crowther’s account apply) as a case where these accounts explain the dependence, novelty, and autonomy of geometric structure (and thus emergence) as it is this specific spatiotemporal aspect that emerges here (because, as noted in Sect. 5.4, a notion of time, another specific spatiotemporal aspect, remains fundamental in quantum graphity).

- (ii) These criteria should accommodate both flat and hierarchical emergence cases into one framework.
- (iii) These criteria should have as clear and precise definitions as possible.
- (iv) These criteria should be as metaphysically neutral as possible.

Why should we adopt these desiderata? The desirability of (i), the avoidance of spatial/temporal/spatiotemporal presuppositions, is obvious, as the QG theories in question in this article are purportedly fundamentally non-spatiotemporal (or at least either fundamentally non-temporal or non-spatial), so presupposing these notions is inappropriate in this context.¹⁷ After all, it hardly makes sense to provide an account of emergence presupposing that spacetime (or either space or time) is fundamental when the non-fundamentality of spacetime is precisely why we are interested in the question of spacetime emergence in the first place.

Desideratum (ii), requiring that we accommodate both flat and hierarchical emergence cases, is of interest, as Crowther (2021) argues that it is possible that spacetime can emerge either flatly or hierarchically. So, our emergence account should be able to accommodate both scenarios. To complicate things further, Crowther (2022) later argues that, in some cases, flat and hierarchical emergence come together, such that both forms of emergence may be present in a single QG context. As such, having a general definition of emergence that accommodates both the flat and hierarchical aspects of the emergence of spacetime is important for this reason, too.

There are two reasons for desideratum (iii), the clarity and precision desideratum. Firstly, clarity and precision are well-established virtues for philosophical positions and definitions (see Williamson (2006), and Stone (2015) for further defense of clarity in philosophy). Secondly, excessive vagueness in our understanding of emergence could lead to situations where it is unclear whether emergence is present or not. As my goal is to find a useful set of emergence definitions well-suited to the spacetime case, a set of definitions that often renders ambiguous results is less useful and more difficult to use. In the context of seeking a generally applicable account of spacetime emergence, the clarity and precision desideratum is particularly needed, as one might wonder what good is an emergence account that may seem to apply to many cases if what it says about those cases is too unclear to be useful? In fact, an unclear account may appear to be more generally applicable than it really is because the apparent plausibility it has in some cases is primarily due to its vagueness. Clarity and precision might seem an obvious requirement, and these are typically implicit norms in conceptual analysis of the kind done here. However, I think making this implicit assumption explicit (in the form of a desideratum) is useful, as it allows us to compare the accounts of emergence discussed in this article based on their relative clarity and precision, and as the article goes on to argue, they can differ in this regard.

As for (iv), the metaphysical neutrality criterion, all that is sought here is a set of helpful definitions that are maximally adaptable to different QG contexts and interpretations. Beyond this relatively narrow issue, the metaphysics of QG is widely

¹⁷I suggest the possibility of theories where either only space or time is non-fundamental, as there may be cases of these, such as quantum graphity (discussed in Sect. 5.4), which, as previously mentioned, retains fundamental temporality.

contested, with few clear conclusions. We should be able to assess potential cases of spacetime emergence and provide criteria for this without getting caught up in such debates. Getting bogged down in metaphysical debates can only restrict the scope of applicability of such criteria and diminish their general utility. As such, metaphysical neutrality is a desideratum, and our emergence definitions should minimise the number of built-in metaphysical presuppositions.

This neutrality comes in two obvious forms. The first is neutrality regarding both the nature of the inter-level relationship in hierarchical emergence and the precise metaphysical nature of emergence itself. Such neutrality is needed as controversy exists both regarding whether the inter-level relation is grounding or supervenience (Lam & Esfeld, 2013; Baron, 2020; Baron et al., 2022), and regarding the status of mereology as a means to understand the inter-level relation (Baron & Le Bihan, 2021; Baron, 2020, 2023). Further debate also exists regarding positions that, rather than providing dependence, autonomy and novelty criteria, seek to explain how emergence metaphysically works in detail. These include role, realiser (Baron et al., 2022; Lam et al., 2018), and constitutive functionalist accounts (Wallace, 2024a), and again, mereology (Baron, 2020).¹⁸

The second form of neutrality is neutrality regarding the ontology of QG itself. This is needed as much debate regarding the ontology of QG theories exists. Here, various structuralist views (Dawid, 2013; Matsubara, 2013; Rickles, 2017), substantialist (Norton, 2020), and priority monist views (Baron & Pohlmann, 2025) are some rival positions. The viability of many staples of metaphysical theorising, such as metaphysical accounts of causation QG (see Sect. 3.2), are also contested in QG. I do not want to prejudge these issues. Instead, I simply seek to provide a maximally useful set of emergence criteria, useful to all parties in such debates (or at least as many as possible).

One might worry that, as emergence itself is a metaphysical notion, demanding metaphysical neutrality is unnecessary.¹⁹ If we grant that emergence is a metaphysical notion, this still doesn't mean that we shouldn't want to avoid building in unneeded controversial metaphysical presuppositions into our account or presuppositions that unnecessarily restrict the domain of applicability of our emergence account, which is significant as I seek a broadly applicable account here. For example, an account of emergence applicable only to one specific (and controversial) interpretation of a theory (e.g., one presupposing a metaphysically robust notion of causation) has a less broad scope than a more neutral account that makes no assumptions about the metaphysics of causation (I develop such an argument in more detail in Sect. 3.2). Similarly, a hypothetical definition of dependence, novelty, and autonomy restricted only to a specific understanding of spacetime emergence, such as the mereological approach, seems less useful than a more neutral account without such restrictions. So metaphysical neutrality seems to lead to a more broadly applicable and thus broadly useful account of emergence.

¹⁸ While largely overlooked in the QG-specific literature, another candidate is real pattern ontology (e.g. Wallace (2024a), Franklin and Robertson (2024)).

¹⁹ This potential objection was helpfully suggested by an anonymous reviewer.

3 Degrees of freedom-based emergence approaches

3.1 Bain's degrees of freedom-based emergence account

With these desiderata established, we can now turn to assessing existing accounts, before presenting a new (at least to this QG context) view. When applying these desiderata, I first examine various views understanding emergence (at least in part) in terms of eliminating degrees of freedom (DOFs). Wilson (2010, 2020) and Bain (2013a, 2013b) provide distinct approaches to this strategy.

Bain's understanding of emergence builds on his broader interpretation of effective field theories (EFTs) in quantum field theory. This is applied to condensed matter analogue models of spacetime emergence, where "spacetime emerges in the low-energy sector of a fundamental condensate" (Bain, 2013b, p. 338). Bain (2013a, p. 6) lists the three key features of his interpretation of EFTs:

- a) Failure of law-like deducibility. If we understand the laws of a theory encoded in a Lagrangian density to be its Euler-Lagrange equations of motion, then the phenomena described by an EFT are not deducible consequences of the laws of a high-energy theory.
- b) Ontological distinctness. The degrees of freedom of an EFT characterize physical systems that are ontologically distinct from physical systems characterized by the degrees of freedom of a high-energy theory [as high-energy DOFs are eliminated when constructing an EFT].
- c) Ontological dependence. Physical systems described by an EFT are ontologically dependent on physical systems described by a high-energy theory.

This leads to an account of emergence where dependence is understood as micro-physical composition, such that "the emergent system should ultimately be composed of microphysical systems that comprise the fundamental system and that obey the fundamental system's laws" (Bain, 2013a, p. 10). The formal distinctness between the effective theory and the base theory implies novelty, with the elimination of DOFs being a key marker of such novelty. No account of autonomy (as distinct from novelty) is given. However, instead of simply taking the formal distinctness of theories as sufficient for novelty, novelty is instead linked to the additional claim that "the properties of the emergent system should not be deducible [or derivable] from the properties of the fundamental system" (Bain, 2013a, p. 10). If an account of autonomy is desired, autonomy in Bain's view could plausibly be underpinned by this non-derivability claim (which is discussed in more detail below) (Crowther, 2015).

Bain's account fulfils desideratum (i), namely avoiding spatial/temporal/spatio-temporal presuppositions, by not presupposing spacetime. The account is strongly tied to the EFT framework for which it is designed, and thus has a built-in commitment to hierarchical emergence (through its commitment to a low-level micro-physical and higher-level/macrophysical picture of emergence) in its definition of dependence, which conflicts with desideratum (ii), requiring that we accommodate both flat and hierarchical emergence. However, it is reasonably clear and precise (fulfilling (iii), the clarity and precision desideratum) as distinct DOFs of the high

and low-energy theories are readily identifiable in the examples Bain (2013a, 2013b) provides. This basic idea of novelty as formal distinctness is ultimately one that I will endorse, albeit stripped of any necessary connection to Bain's derivational independence claim (see Sect. 5 for further details).

However, Bain's view faces issues regarding desideratum (iv), the metaphysical neutrality criterion. Firstly, Bain's view, by emphasising non-derivability, potentially implies a tacit commitment to strong emergence. This is so, as the natural way to describe the strong/weak emergence distinction in QG is that weak emergence refers to the scenario where, when considering emergent E and a base B, E is derivable from B, at least in principle.²⁰ Strong emergence is the view that denies this derivability claim (Baron, 2020). A commitment to strong emergence is a strong metaphysical commitment, which seemingly clashes with (iv) our neutrality criteria.²¹ While I do not rule out the prospect of strong emergence in QG in principle, and discovering that this is present (as Bain's account seems to imply in some contexts) would be interesting. However, the issue with Bain's account is that it seems committed to only strong emergence, which potentially restricts the domain of applicability of this account. Ideally, a maximally metaphysically neutral (and most broadly applicable) account should be able to accommodate both strong and weak emergence. Franklin and Robertson (2024) imply that their view is neutral regarding weak and strong emergence.²² There are also various formulations of Wilson's (2020) DOF-based definition of emergence applicable to both strong and weak emergence, so both these accounts seem more generally applicable regarding this issue than Bain's view.

Additionally, Crowther (2015, 2016) argues that defining emergence in terms of the failure of reduction, such as by incorporating non-derivability into one's definition of emergence, requires us to take a strong stance on the nature of derivability and reduction, and opinions vary on this issue.²³ As Crowther (2015, p. 427) describes, Bain's derivational independence claim is based on "the fact that, in practice, the use of approximations and/or other means of external guidance is ... [typically] ... necessary ... to construct the low-energy EFT, even given the high-energy theory", where the use of such methods implies a failure of lawlike deducibility, and do not count as derivations. So, Bain's view potentially violates desideratum (iv), the metaphysical neutrality criterion, if we take this stance on derivation to have metaphysical import, as Bain does.

²⁰ Definitions of strong emergence in terms of novel causal powers (e.g. Wilson, 2020) or downward causation are inappropriate in QG, as the viability of causal notions in QG is contested (as Sect. 3.2 argues).

²¹ Ellis (2020) also identifies Bain's view as potentially implying strong emergence.

²² Franklin and Robertson (2024, p. 14) claim that their view is compatible with both weak and strong emergence, stating that "we believe that inter-theoretic reduction is plausibly widespread, it is not a requirement of our account" and that their "account is compatible with higher level entities' being strongly emergent". However, Franklin and Robertson (2024, p. 14) also note that "strong emergence shouldn't be a foregone conclusion of our account" as their view is compatible with inter-theoretic reduction and derivability and thus weak emergence, which their article mostly focuses on. We should also note that Crowther (2021, 2022), makes no explicit statement in favour of strong or weak emergence, so may also be neutral regarding this.

²³ Bain's derivational independence claim is not uncontroversial even in QFT, as Crowther (2015) notes. See Rivat and Grinbaum (2020) for further discussion.

Moreover, the derivability of GR from a QG theory is a generally accepted requirement for a viable QG theory (Crowther, 2016, 2018). Thus, defining emergence through derivational independence is a risky move in QG, as Crowther (2015, 2016) points out. One's stance on this may, of course, depend on what one means by deriving GR from QG. If it means something akin to the reconstruction of GR as an EFT, one could debate whether this counts as a derivation.²⁴ I'm not trying to resolve the debate about what constitutes a derivation in this context here. Instead, I will simply note that Bain's view takes a particular stance on this (which is then taken to have metaphysical importance, due to its connection to the above strong emergence claim and its centrality to his definition of emergence), in a manner that seemingly conflicts with the neutrality desideratum (iv). Thus, while I will ultimately accept Bain's view of novelty as formal distinctness, the somewhat contentious metaphysical commitments implied by the other aspects of his view restrict the scope of applicability of the account in the context of spacetime emergence in QG.

3.2 Wilson's DOF-Based emergence

Having discussed Bain's view, I will now consider Wilson's (2013, 2015, 2020), view. In Wilson's view, an entity *E* is emergent if it possesses a proper subset of the DOFs of its base *B*. Wilson (2013) contrasts this with restrictions and reductions of DOFs, which are not considered cases of emergence.²⁵ The elimination of DOFs is understood as the elimination of causal powers. We can then define dependence and autonomy in these terms.

Dependence: An emergent *E* "is composed of the [set of base entities B_i], as a result of imposing some constraint(s) on the" base (Wilson, 2010, p. 292). Wilson (2020) later characterises such dependence as cotemporal material dependence.

²⁴ I would like to thank an anonymous reviewer for highlighting this point.

²⁵ For Wilson (2013, pp. 224–225) "[f]irst, constraints ... reduce the DOF needed to characterize an entity as being in a given state ... [e.g.] ... a point particle constrained to move in a plane has 2 configuration DOF, rather than the 3 configuration DOF needed to characterize a free ... particle ... Second, constraints ... [can] ... restrict DOF.[e.g.] ... A point particle may be constrained ... to some region including and above the plane ... Third, sometimes ... constraints eliminate ... DOF ... [e.g.] ... N free point particles, having $3N$ configuration DOF, might come to compose an entity whose properties and behaviour can be characterized using fewer configuration DOF ... because the properties and behaviour of the composed entity are functionally independent of these DOF". Only the latter cases count as emergent in Wilson's (2010, pp. 293–308) view, because constraints which "only reduce or restrict the DOF needed to characterize [and entity] *E*, are arguably ontologically reducible to their composing e_i [their base entities, because] ... unlike mere reductions or restrictions in DOF, [eliminations] completely eliminate functional dependence of the properties and behavior of the composed entity on the parameter at issue".

Autonomy: As E has fewer DOFs and thus fewer causal powers than B, B cannot be identical to E via Leibniz's Law (Wilson, 2020). This implies E is autonomous from B, due to the difference in DOFs and causal powers.²⁶

Wilson (2010, 2020) gives no separate definition of novelty. If one sees this as problematic, then one of the other novelty criteria discussed elsewhere in this article could be added to Wilson's view. Alternatively, we could view novelty as also defined by the core aspect of the autonomy criteria, namely the difference in causal powers between E and B, giving:

Novelty: An emergent E is novel because it possesses a non-identical set of powers compared to its base B.

Crowther (2018 p. 84) briefly lists some applications of Wilson's view, claiming that:

[it].is ... applicable if spacetime emerges as illustrated by the condensed matter approaches to QG, and ... applies to GFT [group field theory], and ... other approaches where spatiotemporal ... [DOF] ... emerge as collective, low-energy variables, analogous to those of thermodynamics. It also may apply in ... LQG, where degrees of freedom possessed by the spin foams are eliminated in the approximation and limiting procedures designed to resolve and/or wash-out their discrete nature and quantum properties in the recovery of spacetime.

How does this fare regarding our desiderata? Wilson's (2020, p. 72) general understanding of weak emergence presupposes "cotemporal material dependence", where cotemporal dependence is the dependence of E upon B at a particular time. This violates desideratum (i), the avoiding spatiotemporal presuppositions-related desideratum, which requires that our emergence criteria do not presuppose spatiotemporal notions. This can be fixed by replacing cotemporal dependence with a more general notion of hierarchical dependence, where our emergent E depends on a base B, with this inter-level dependence understood via a non-temporal conception of a hierarchy of levels. So, this is a problem, but a fixable one.²⁷

I also see no reason why a view based on eliminating DOFs and causal powers is necessarily incompatible with flat emergence, as a transformation of a system occurring at the same level seems, in principle, capable of leading to the elimination of DOFs in the emergent state E when compared to B, such that E has non-identical causal powers than B. Therefore, desideratum (ii), which requires compatibility with

²⁶This is Wilson's account of weak emergence. Wilson also gives an account of strong emergence, which combines cotemporal material dependence with E's possession of causal powers absent in B (Wilson, 2020). However, for the reasons previously discussed, strong emergence is not the main focus here, even if I do not rule it out. Moreover, by defining strong emergence in term of causal powers, Wilson's strong emergence view also conflicts with the metaphysical neutrality desideratum (iv).

²⁷Something like this seems tacitly presupposed in Crowther's (2018) aforementioned discussion of Wilson's view.

both flat and hierarchical emergence, appears to be fulfilled.²⁸ Like Bain's view (iii), the clarity and precision desideratum, is seemingly fulfilled as the notion of the elimination of DOFs is reasonably precise and identifiable in the cases under consideration here.²⁹

Crucially, when considering Wilson's view in QG, Crowther (2018) does not mention the core of Wilson's approach: its commitment to causal powers. This metaphysical commitment obviously violates the neutrality criterion (iv). Wilson's view is neutral regarding the precise metaphysics of causation, and purportedly accommodates Humeanism-friendly Lewisian counterfactual views, although Ney (2022) expresses skepticism regarding this. However, Wilson (2020, p. 34) explicitly commits to the "assumption of physical causes and powers" and explicitly rejects Russell's (1913) causal eliminativism. Neo-Russellianism instead denies the applicability of causal notions (and causal metaphysics) to fundamental physics, even if causation is useful in the special sciences. Thus, causation is understood to exist non-fundamentally but not fundamentally.³⁰ As Wilson's view clearly requires that we are able to speak of both fundamental and emergent causal powers, it also rules out Neo-Russellianism. These are strong metaphysical commitments clearly violating (iv), the metaphysical neutrality desideratum.

While I am not arguing that Neo-Russellianism is true, ruling this out is unwise, as Ney (2022) and Blanchard (2016) identify the prominence of Neo-Russellianism among philosophers of physics. Frisch (2023) highlights five basic arguments Neo-Russellians invoke as grounds for scepticism of causation.³¹ Should any of these succeed regarding GR, then we cannot even characterise emergent spatiotemporal DOFs in terms of causal powers. Neo-Russellianism potentially gains additional support because many approaches to the metaphysics of causation typically presuppose spatial, temporal or spatiotemporal notions to be viable, including dispositionalism, causal process theories and Lewis's counterfactual views, according to Baron et al.

²⁸ We should note that Wilson (2024) expresses scepticism regarding diachronic emergence. This plausibly extends to flat emergence. Wilson argues that many putative cases of diachronic emergence are actually cases of causation. However, invoking the notion of causation is problematic for QG for the reasons discussed below, so Wilson's argument might not be applicable in QG.

²⁹ The one exception is that Wilson's view may yield ambiguous results when applied to our quantum graphity case. It is unclear to me whether the switching off of the links in quantum graphity restricts or eliminates DOFs. Only the elimination of DOFs (not restriction or reductions of DOFs) implies emergence for Wilson.

³⁰ See Price and Corry (2007) and Blanchard (2016) and the references therein for a discussion of neo-Russellianism, and Ladyman and Ross (2007) and Farr and Reutlinger (2013). For a recent neo-Russellianism with a pragmatist twist, see Rovelli (2023).

³¹ These are the vagueness, dominant cause, determinism, locality, and time asymmetry challenges to causation in physics (Frisch, 2023).

(2022).³² Purportedly QG-friendly counterfactual causation approaches exist (e.g. Baron & Miller, 2015; Baron et al., 2022), but are controversial (see Tang, 2023).³³

Instead, the most tenable form of causation in QG may be interventionist (e.g. Baron & Le Bihan, 2024).³⁴ However, Baron and Le Bihan (2024, p. 21) concede that the type of dependence they utilise “might not be considered as causal by everyone”. More generally, whether interventionism amounts to a metaphysics of causation is contested, and it is often argued that interventionism is actually compatible with neo-Russellianism (Blanchard, 2016; Woodward, 2007).

Given the issues Wilson’s view faces regarding desideratum (iv), the metaphysical neutrality criterion, it is useful to look elsewhere for a generally applicable account of emergence in QG.

4 Crowther’s view

Crowther (2021, 2022) gives several emergence definitions, where E represents an emergent and B its base.³⁵ Crowther begins with a general definition of emergence, where E depends on B, if E is derivable from B, or if E supervenes on B. Novelty is then defined as E being “strikingly, qualitatively different from B” (Crowther, 2022, p. 386).³⁶ Autonomy is defined as E’s robustness regarding changes in B. In other words, E is autonomous if it is the same for a number of assumptions regarding, or choices regarding “comparison class, B” (Crowther, 2022, p. 386).

More specific versions of this basic idea are given for flat and hierarchical emergence. In the hierarchical case, dependence is understood in terms of the physics of the more coarse-grained theory E being supervenient on that of B or derivable from B. Autonomy is defined in terms of a macro description E’s robustness with respect to a comparison class of micro details B, and novelty as the macro theory differing remarkably from the micro theory. For flat emergence, the talk of fine-grained and coarse-grained theories and macro and micro theories (both of which imply inter-

³² Lam et al. (2023) suggest the possibility of a non-spatiotemporal dispositionalism (albeit in the context of debate regarding laws, not causation), but do not find this unproblematic.

³³ For an objection to Baron Miller and Tallant’s view, in the context of T-duality in string theory, the common core interpretation (which identifies the ontology of string theory with the duality invariant common core) implies that we cannot treat the dimensionality of target space or winding space (the logical candidates for a structure providing a notion of location) as a part of the physical common core (Huggett & Wüthrich, 2025). Ultimately, “T-duality ... implies that the same phenomenology can be described in terms of theories which differ in size, momentum, topological position and even the dimensionality of their objects” (Dawid, 2013, p. 181). This casts doubt on the notion of the topological position of objects, as we lack a clear account of their location in the fundamental QG structure. Yet, Baron, Miller and Tallant’s (2022) proposal for fundamental counterfactual-based causation presupposes the possession of a clear notion of location in a QG structure.

³⁴ However, Le Bihan’s (2020, p. 17) claim that LQG implies the “denial of the partial ordering of events” potentially challenges causation accounts presupposing this, potentially including interventionism.

³⁵ Here, I focus on the most recent expression of Crowther’s view (Crowther, 2021, 2022), which streamlines the core insight of Crowther (2016) and accommodates flat and hierarchical emergence.

³⁶ It is worth noting that Crowther (2016) earlier endorsed a definition of novelty in terms of formal distinctness akin to the view I endorse Sect. 2, but later dropped this in Crowther (2021, 2022). It is this latter view that I consider in this section.

level or hierarchical emergence) is dropped, but the definitions remain similar. In flat emergence dependence is defined as flat supervenience, such that “E flatly supervenes on B (Flat supervenience means there would be no change in the E state unless there were a change at the B state, but not vice versa)” (Crowther, 2021, p. 7296). E’s robustness regarding changes in B, such that the ‘prior’ state B is underdetermined by E, defines autonomy, and novelty is characterised by the somewhat vague notion that E and B remarkably differ (but no mention of levels is made) (Crowther, 2022).

Crowther’s view satisfies three of the desiderata, as it does not presuppose space-time, and Crowther (2021, 2022) gives numerous examples of its applicability to flat (e.g. quantum graphity) and hierarchical emergence (e.g. LQG) QG cases, fulfilling (i), the avoiding spatiotemporal presuppositions desideratum, and (ii), the accommodating flat and hierarchical emergence desideratum. It is also reasonably metaphysically neutral, as desideratum (iv), the metaphysical neutrality criterion, requires.³⁷ However, due to the vagueness of some of these claims, particularly the definition of novelty, Crowther’s view has issues fulfilling desideratum (iii), the clarity and precision desideratum.

Regarding (iii), our clarity and precision desideratum, the definition of novelty in all of Crowther (2021, 2022) definitions is problematic, as the notion of ‘remarkable’ or ‘qualitative’ difference is vague (and neither the terms remarkable nor qualitative are given definitions). As a result, it is unclear what the intersubjective standards for what counts as ‘remarkable’ and ‘striking’ are, and it is plausible that what is con-

³⁷ Although if one does not view supervenience as a metaphysically innocuous notion, the framing of the first and the third of Crowther’s definitions of dependence in terms of supervenience could be seen as conflicting with desideratum (iv), the metaphysical neutrality criterion. Huggett and Wüthrich (2025, p. 173) note that the idea “GR supervenes on a ... [QG theory] ... such as LQG just in case all its predicates supervene on the predicates of LQG for a fixed set ... of objects on which the predicates of both GR and LQG are defined ... runs into the obvious problem that by requiring a fixed set of objects, the characterization only finds application for a relation between two theories sharing a joint ontology of objects on which the ideologies of the theories are then defined”. It is then noted that there is no guarantee whatsoever that the ontologies of LQG and GR match up in this manner, and that even on plausible attempts to relax this requirement for a matching object-ontology, as “GR[’s ontology] ... does not merely consist in composites of the ontology of LQG ... it ... remain[s] ... unclear how the predication induced by LQG on GR’s ontology would work” (Huggett & Wüthrich, 2025, p. 173). This also implies that characterising supervenience in metaphysically neutral terms without taking a stance on the ontology of LQG may be difficult. If supervenience, as Huggett and Wüthrich understand it, presupposes an object-based ontology (as opposed to, say, a process ontology), this is itself not metaphysically neutral. However, I will not press this point in further detail here. Interestingly, Crowther (2015, 2016), as discussed in Sect. 3, cautions against using the notion of derivability when defining emergence, in part because what counts as a derivation is contested and unclear (with this unclarity itself potentially being an issue regarding the clarity and precision desideratum (iii)). Yet Crowther (2021, 2022) returns to utilising derivability when defining emergence. As previously mentioned, a natural way to think about the weak and strong emergence distinction is to define strong emergence in terms of the failure of derivability. So, building the notion of derivability into one’s definition of emergence might imply a rejection of strong emergence (defined via non-derivability). I already argued (in Sect. 3.1) that, ideally, our emergence definition should be neutral between strong and weak emergence, so there might be an issue regarding desideratum (iv), the metaphysical neutrality criterion, here too.

sidered surprising could vary significantly depending on the background of the one making the judgment.³⁸

Moreover, one could claim that we cannot recognise qualitative novelty (of the kind Crowther is cornered with) simply by examining the structure of the theory. This is problematic for Crowther's account, to the extent that this account is intended (as it is) to describe the emergence of theories or the emergence of aspects of theories, instead of just phenomena. An appeal to qualitative novelty plausibly just establishes the novelty of the phenomenon. As such, it doesn't seem precise enough as a means of evaluating inter-theoretic emergence.³⁹ Thus, Crowther (2021, 2022) definition of novelty appears to be insufficiently clear and precise.

While not explicitly stated like this, Crowther specifies autonomy in counterfactual terms. E is robust against changes in B if it is robust under various counterfactual scenarios, like a variation in the configuration of B, or even wholly different underlying theories of B.⁴⁰ Crowther (2021, 2022) notion of robustness is on the right track, as the numerous examples she considers demonstrates. However, it can be improved. One way to do this is to render some of the notions that Crowther uses formally precise, so that we can apply these with greater ease and precision.

The conditional probability-based approach, which I outline in the next section, allows us to upgrade our dependence and autonomy definitions from definitions provided in ordinary language to more precise formalised probabilistic definitions. The conditional probability-based approach also suggests an alternative strategy for defining novelty (and one akin to a view earlier endorsed by Crowther (2016), unlike Crowther's later (2021) and (2022) positions), that avoids the above issue with Crowther (2021, 2022) views with regards to desideratum (iii) the clarity and precision desideratum.

5 The conditional probability-based approach to emergence

5.1 Introducing the conditional probability-based approach

This section introduces the *conditional probability-based approach* to emergence in QG. I argue that this fulfils all the desiderata. Franklin and Robertson (2024)

³⁸This is a problem that Franklin and Robertson (2024, pp. 13–14) raise as an objection to Butterfield's (2011) view of novelty as surprising relative to a comparison class, as they put it: "the differing metaphysical tastes of scientists and analytic metaphysicians might impact their judgements about novelty, rendering the broader enterprise circular. Their differing tastes might themselves be explained by features of the judgments they have been exposed to. For instance, philosophers trained on logic judge that if the laws of fundamental physics are complete, then everything else 'follows from' this – not leaving much room for novelty. In contrast, physicists witness radically different behaviour in different domains" and would likely disagree with the analytic metaphysician and be more generous regarding judgments of novelty. Such an objection plausibly extends to Crowther's similar view of novelty.

³⁹I thank an anonymous reviewer for suggesting this point.

⁴⁰This is identified via either the underdetermination of the microstate (or model) of the theory or via underdetermination of the micro-theory itself.

and Franklin (2024) provide the original conditional probability-based approach to dependence and autonomy.⁴¹

For Franklin (2024, p. 291):

[E]ntities or kinds emerge if and only if they are involved in dependencies that are novel and screen off lower-level details [LLDs]. Screening-off is understood as the combination of un-conditional relevance and conditional irrelevance ... thus, our screening-off criterion entails that the higher-level dependencies are both dependent and independent [autonomous] of the lower-level.

So, when considering a macrodependency C-D that arises from some underlying LLDs, screening-off involves:

- (1) Unconditional relevance: conditional on a particular lower-level description, the probability of the macro-description D obtaining increases: $P(D|LLD) > P(D)$.
- (2) Conditional irrelevance (full generality): $P(D|C \& LLD) \approx P(D|C) = x$ where $0 \leq x \leq 1$ (Franklin, 2024, p. 291).⁴²

For example, following Franklin and Robertson (2024), consider a kettle, where C represents the temperature, and D the boiling, and the LLDs are the molecular motion of the water molecules. The water molecules' molecular motion is unconditionally relevant to the boiling of the kettle because "if they are moving faster, then the kettle is more likely to boil", so the probability of D given the LLDs increases (Franklin & Robertson, 2024, p. 8). Thus, we can plausibly say that the boiling depends on such LLDs. However, once we conditionalise on water temperature C, "any further details of the molecular motion are irrelevant", fulfilling (2) (Franklin & Robertson, 2024, p. 8). Consequently, it is plausible to think that D is autonomous from the conditionally irrelevant LLDs. Thus, (1) can be identified as a dependence condition and (2) an autonomy condition.

Another example is the ideal gas law. The underlying quantum state and quantum dynamics are unconditionally relevant, with the idea that "if matter were not quantum, it wouldn't be stable" used to motivate this dependence claim (Franklin & Robertson, 2024, p. 12). However, the (autonomous) ideal gas law screens off such conditionally irrelevant LLDs, and pV/nR determines the temperature (Franklin & Robertson, 2024, p. 12). Franklin (2024) gives another example where it is argued that quasi-classical worlds in the Everett interpretation are both dependent on the (unconditionally relevant) underlying quantum LLDs, yet, for Franklin, due to decoherence, such LLDs are argued to be conditionally irrelevant such that emergent quasi-classical worlds display autonomy.

⁴¹ Here, in line with my adoption of a neutral stance regarding metaphysical and theoretical emergence, I also remain neutral regarding the nature of probability (e.g. is it epistemic or ontic), and neutral regarding debates in the philosophy of probability more generally.

⁴² Franklin and Robertson use the letters B-A to represent this macrodependency, but since I used B to represent the base below, I made a simple substitution here to prevent confusion.

The next question is whether Franklin and Robertson's and Franklin's approach can accommodate both flat and hierarchical emergence. The use of the term LLDs explicitly confirms that their view, in its original form, is a hierarchical emergence definition. We therefore need something more general that is applicable to flat and hierarchical emergence.

If we consider the relationship between any emergent relationship or dependency (of states, structures and/or entities) E_1 - E_2 , and a set of base details B , I propose the following definitions:

- (1) **Dependence** is defined as unconditional relevance. If we conditionalise on some base details/physical configuration B , the probability of the emergent description E_2 increases: $P(E_2|B) > P(E_2)$.⁴³
- (2) **Autonomy** is defined as conditional irrelevance: $P(E_2|E_1 \& B) \approx P(E_2|E_1) = x$ where $0 \leq x \leq 1$.⁴⁴
- (3) **Novelty** is identified with a formal (often dynamical) distinctness.

Here, we can regard our base B as a set of lower-level details (in hierarchical emergence) or a 'prior' state of the system (on the same level) in flat emergence. Moreover, we can think of (2) as a useful clarification and enhancement of Crowther's notion of robustness, as what it is for our emergent E to be robust with regards to changes in B (in both flat and hierarchical emergence) is for the details of this base to be approximately conditionally irrelevant where our emergent description (GR) is known to be well-tested (with conditional irrelevance formally defined via (2)), such that this emergent general relativistic spatiotemporal picture can be understood as autonomous.

Regarding novelty, I follow a common trend in the literature that identifies this with formal (and often dynamical) distinctness. We have already met one such definition, Bain (2013a, 2013b). Franklin (2024), Franklin and Robertson (2024), and (Crowther, 2016), in contrast to her later 2021 and 2022 positions discussed previously) adopt similar views. Several key markers of such formal distinctness are evident in the literature. They include changes in variables, as Franklin (2024), drawing on Knox (2016) and Franklin and Knox (2018) notes, alongside differences in DOFs,

⁴³This view can accommodate an equivalence class of base details, $B_1, B_2 \dots B_N$ for which $P(E_2|B_i) > P(E_2)$. Thus, it can accommodate multiple realizability (Franklin & Robertson, 2024).

⁴⁴This conditional irrelevance need only be approximate, as situations exist where the base details do not always develop into the base details underpinning our emergent state (Franklin & Robertson, 2024).

as highlighted in Bain's aforementioned view, and Crowther (2016). Crowther (2015) also identifies distinct symmetries as relevant formal distinctness.⁴⁵

In Franklin and Robertson (2024) account, novelty can be inferred via multiple means, including nomic distinctness, dynamical distinctness, or causal distinctness. Of the three forms of distinctness mentioned here, Franklin (2024, p. 292) concentrates on dynamic distinctness (and the possession of distinct functional form) in mathematical physics, as in "mathematised sciences, the realisation of distinct dynamics [with distinct functional form] is sufficient for novelty". My account broadly follows Franklin's move here. Moreover, Franklin and Robertson (2024) themselves note that invoking causal novelty is potentially problematic in fundamental physics. Additionally, defining novelty in terms of causation clashes with (iv), the neutrality desideratum.

My approach, which defines novelty in terms of formal distinctness, is thus in the same spirit, but it is more general than Franklin's approach, as it can accommodate other relevant forms of distinctness. I use the term formal distinctness (also used by Bain (2013a, 2013b), and Crowther (2016)) because, in some cases in QG, the kinematic structures of the theories are more well-developed than their dynamical structures (such as in canonical LQG). However, in this case, there still appears to be a notion of novelty here between the kinematic structure of LQG (as a quantum theory) and any classical spacetime theory. Franklin (2024) takes the difference between classical and quantum theories to be an obvious example of novelty. This plausibly remains the case regarding QG and classical spacetime theories. Such novelty can plausibly be captured by the notion of formal distinctness, which is broad enough to accommodate both relevant dynamical and kinematic distinctness/novelty. Additionally, while the kinematics/dynamics distinction is prominent in many current QG approaches, Lam and Wüthrich (2023 p. 21) suggest that "this distinction cannot be presupposed to obtain or ... remain meaningful in ... quantum gravity". So, speaking of formal distinctness seems more perspicuous, as it can accommodate this scenario.

⁴⁵ Franklin (2024, p. 292) also notes "that while one might concoct borderline cases of distinct functional form, all would likely agree that the dynamics of classical and quantum mechanics qualify as sufficiently distinct". Similarly, the formal distinctness of classical general relativity (GR) and its QG base is also clearly evident in the examples considered here. So, we needn't worry about cases of gerrymandered formal distinctness, or formal distinctness between different formulations of the same theory operating at the same level, or between dual theories (as we don't want to count the distinctness between the ADS and CFT sides of ADS/CFT duality to be novel in an emergent sense, since this is not typically understood as a case of emergence (Crowther, 2016)). If we wanted to be more precise about the latter case, we could say that the formal distinctness relevant to emergence involves the failure of a duality map, although I will not develop this suggestion in detail here.

The notion of nomic distinctness can also fall under the broader umbrella of formal distinctness as understood here.⁴⁶

Thus, we have a definition of novelty independent of the use of conditional probability for defining dependence and autonomy. Nevertheless, for simplicity's sake, I will treat this novelty definition as part of a broader framework for understanding emergence, referred to here as *the conditional probability-based approach*.

How does the above view of emergence fare regarding our desiderata? As I have not mentioned spacetime or spatiotemporal location in the definition of this view, it does not explicitly presuppose spacetime, fulfilling (i), the desideratum requiring no spatiotemporal presuppositions. Nor does it implicitly presuppose spacetime, as conditional probability and formal distinctness are definable independently of spatiotemporal physics or concepts. Conditional probability is a mathematical notion that is independent of spacetime physics, and formal distinctness can exist between any theoretical structure, regardless of whether they are spatiotemporal or non-spatiotemporal.

The account of emergence I have offered fulfils desideratum (ii) the accommodating flat and hierarchical emergence desideratum, by accommodating both flat and hierarchical emergence within one account, as our base B could either be a set of lower-level details (in hierarchical emergence), or a 'prior' state of the system (on the same level) in flat emergence. Employing clear and precise formal definitions stated in terms of conditional probability plausibly fulfils (iii), the clarity/precision desideratum, regarding dependence and autonomy. Defining novelty as formal distinctness is reasonably clear and precise, even if this notion is not as readily formalisable in probabilistic terms as (1) and (2). Moreover, key means of identifying such distinctness have been presented (distinct DOFs, variables and symmetries). As no metaphysical concepts are mentioned in any of the above definitions, they are highly metaphysically neutral, fulfilling (iv), the metaphysical neutrality desideratum. So, to sum up our results, we have an approach to defining dependence, autonomy, and

⁴⁶I thank an anonymous reviewer for encouraging me to elaborate on these formal distinctness related issues. The reviewer also notes that Bain's notion of formal distinctness could be considered a form of nomic distinctness (such as cases involving Lagrangian densities, for example). I agree, and consider the idea of formal distinctness to account for differences in the structure of theories, whether they are kinematic, dynamic, or nomological. However, we might not want to exclusively privilege nomological distinctness, as Lam and Oriti (2024) raise the possibility of an anti-realist view of laws in QG (a view I am neutral on here), so using something more general (the notion of formal distinctness) that doesn't rule this out is useful. Franklin and Robertson (2024, p. 14) also note that "[a] macrodependency is novel ... if it's not type-identical to the microdependencies that instantiate it." However, this notion may face issues regarding desideratum (iv) if identifying types involves metaphysical commitments, although I will not explore this in detail here, due to limited space, and as Franklin (2024) similarly omits discussion of type identity.

Table 1 Comparison of the four emergence approaches with respect to the desiderata

Desiderata	Bain's view	Wilson's view	Crowther's view	Conditional Probability-Based Approach
i. Non-spatiotemporal	☑	☑	☑	☑
ii. Flat and hierarchical	☐	☑	☑	☑
iii. Clear and precise	☑	☑	☐	☑
iv. Metaphysically neutral	☐	☐	☑	☑

novelty that fulfils all our desiderata (see Table 1 for a table overviewing this), making it the most generally applicable account on offer.⁴⁷

5.2 Applications of the conditional-probability-based approach: LQG

Now I will demonstrate how the conditional-probability-based approach to emergence works by providing some examples. So, let us first consider LQG, a potential hierarchical emergence case. I will focus on canonical LQG here. In this context, recall the notion of spin-networks and weave states discussed in Sect. 2.1.

First, I will apply the dependence/unconditional relevance criterion. In LQG, the base details, namely the spin network structure, are unconditionally relevant to the existence of any emergent classical spacetime and its behaviour (fulfilling (1), unconditional relevance). Here, we are evaluating the conditional probability of states approximating classical spacetime in LQG.⁴⁸ Within the LQG framework, if we conditionalise on some LQG base details (or equivalence class of these), this base is unconditionally relevant because we could not have spacetime (and spatiotemporal area and volume) in LQG without it. To illustrate this, consider that the probability of recovering classical area and volume in the standard formulation of canonical LQG without spin networks in weave states (see Sect. 2.1, and equations (1) and (2) therein for an overview of weave states) is plausibly zero.⁴⁹ So, the presence of spin networks in weave states obviously increases the probability of an emergent spatiotemporal (i.e. spacetime approximating) description in LQG, as the probability

⁴⁷ It is, however, worth noting that the advantages conditional probability-based approach possesses over Crowther's view are more subtle than the other two cases and largely hinge on the issues Crowther's definition of novelty faces that are discussed in Sect. 4, as well as the extra precision gained by the formalisation of the definition of autonomy and dependence. While Crowther's account and the conditional probability account don't disagree on existing cases, a clearer definition of novelty is also beneficial when considering new cases, as it will help us assess them more easily. One could even claim that its greater precision helps us conceptualise emergence, as we can more easily see what's ruled in and what's ruled out (particularly regarding novelty). Additionally, footnote 37 raises potential issues with both aspects of Crowther's definition of dependence. To the extent that these are convincing, this could be an additional advantage of the conditional probability-based approach, which avoids these issues by being neutral regarding the status of supervenience and derivability when defining dependence.

⁴⁸ We should note that the semiclassical limit of LQG, and its capacity to recover GR more generally is a topic of ongoing research, so any claims about GR's recovery are provisional on the success of this (see Rovelli and Vidotto (2014) for discussion).

⁴⁹ Following Huggett and Wüthrich (2025), for the purposes of this discussion of emergence in LQG, I set aside other related approaches to approximating spacetime, such as coherent states, or 'shadow states' (See also Thiemann (2007) for discussion of these).

of spacetime being present without weave states is zero in canonical LQG, but it is nonzero with them. Thus, any emergent spacetime in LQG plausibly depends on the presence of such weave states (and thus on the unconditionally relevant spin-network base details). Thus, if E_2 represent a spatiotemporal description (i.e. one approximating classical spacetime) in LQG and B the LQG base details, the probability of the emergent description E_2 increases: $P(E_2|B) > P(E_2)$. This establishes that any emergent spacetime in LQG depends on the base details.⁵⁰

Now I will apply the autonomy/conditional irrelevance criterion. There are features of LQG that are plausibly conditionally irrelevant at low energies, where GR is to be recovered. If we consider disordered locality, for Wüthrich (2017, p. 316):

These non-localities are suppressed in the low-energy approximation from the spin network to the relativistic spacetime ... they must be suppressed, for otherwise they would have to be emulated by the emerging structure in the sense that these adjacency relations would re-occur in the spacetime in the form of neighbouring relations and thus not qualify as ‘non-localities’ ... If the course graining attendant to the emergence of spacetime from spin network states ... would not ‘wash out’ the non-local connections, they would have to be encoded in the emerging relativistic spacetime, perhaps as non-local ‘worm-holes’. If, however, their presence were so strong as to preclude essentially local physics at comparatively low-energy scales ... then the corresponding theory, or ... model, would ... be considered empirically inadequate.

Consequently, the disordered locality of LQG must be screened-off (and approximately conditionally irrelevant) at scales where a classical non-locality-free general relativistic description of spacetime applies, even if the precise details of this and the recovery of GR from LQG are still under development.⁵¹ This seemingly implies that the general relativistic picture we recover from LQG is autonomous (in the relevant sense defined in Sect. 5.1) from the LQG base details that display disordered locality (as such disordered locality is screened off at macroscopic scales where GR is well tested).

We can further clarify this autonomy claim. Our ability to use GR effectively (via providing useful predictions and explanations within its regime of empirically verified applicability), without consideration of the LQG base details in regimes where GR is well-tested and known not to break down, serves as evidence of the conditional irrelevance of such LQG base details in these contexts. This is because our use of

⁵⁰ While I have so far focused on weave states in the canonical approach, in the covariant approach, for Rovelli and Vidotto (2014, p. 162), “Coherent states... connect the quantum theory ... with classical general relativity.” So, the story of GR’s recovery from LQG involves “coherent states in the Hilbert space of the theory, and their use in proving the large-distance behaviour of the vertex amplitude and connecting it to the Einstein equations” (Rovelli & Vidotto, 2014, p. 162). In this context, we also would not have our emergent spacetime without the relevant LQG base details on which the emergent spacetime depends, plausibly fulfilling (1).

⁵¹ For brevity’s sake, I will not go into detail, but the generic quantum features of LQG, such as superposition, are also plausibly screened-off, for GR to be recoverable. Even LQG’s fundamental discreteness itself must be conditionally irrelevant at the scales where GR is well-tested for LQG to approximate/recover GR successfully.

GR in macroscopic regimes (e.g. when calculating the precession of the perihelion of Mercury), without consideration of the underlying LQG base details, would not work if such LQG base details are not approximately conditionally irrelevant. So, the precise quantum, low-level, LQG base details (including those subject to disordered locality) are conditionally irrelevant when we conditionalise on a particular macro-state E_1 (when using GR to describe a macroscopic relationship between spatiotemporal states E_1 - E_2 that we calculate using GR, such as the precession of the perihelion of Mercury calculation) when applying our autonomy criterion (2).⁵²

As a result, it is plausible to consider our emergent classical picture of space-time as autonomous (in the sense defined in Sect. 5.1) from the underlying LQG details (or, in Franklin and Robertson's terminology, the LLDs). Thus, criterion (2), the autonomy criterion, is also plausibly fulfilled, and our emergent state (that can be modelled via GR) is autonomous from the base details within LQG.

Having established the autonomy of our emergent classical description of space-time, we can now consider this emergent classical picture (as it is described by GR) to evaluate novelty. Here, novelty is trivially satisfied (regarding LQG's kinematics) given the obvious formal distinctness between discrete quantum spin-network structure, with distinct quantum variables, and DOFs from classical GR.⁵³ LQG's dynamics are still a work in progress.⁵⁴ If we just consider the spin-foam approach to covariant LQG, we also have quantum spin-foam dynamics clearly formally distinct from GR (and differ regarding their high-energy DOFs that are eliminated at low energies in the recovery of GR, as previously mentioned by Crowther (2018) in Sect. 3.2).

5.3 String theory

It's also interesting to see how the conditional probability-based approach fares in string theory. Simply put, in string theory, the universe is understood in terms of one-dimensional strings and has more dimensions than are readily apparent (10 in current superstring theories). Observed particles are vibrational modes of these strings, and gravitational interactions are carried out by a particle called the graviton, which is identified with a specific low-energy vibrational mode of a closed string. Modern string theory is a framework comprising five types of string theories related via dualities (Zwiebach, 2009). There is no clear consensus about the interpretation of dualities. The common core approach (where the physical content of string theory is identified with the duality invariant common core) is popular (Le Bihan & Read, 2018), but has notable critics, including Read et al. (2020) and Grimmer D Cinti et al. (2024).

According to Huggett and Wüthrich (2025, 330):

⁵² Thus, the condition $P(E_2|E_1 \& B) \approx P(E_2|E_1)$ is fulfilled if B represent our LQG base details.

⁵³ As "spin networks can be seen to represent new degrees of freedom, rather than simply a quantisation of the continuum fields of GR" (Crowther, 2021, p. 7294).

⁵⁴ Notably, the Hamiltonian constraint that encodes its dynamics, lacks a complete solution, an issue particularly troublesome for the canonical approach.

- (i) ... suitable string states correspond to quanta of the gravitational field, and
- (ii) ... such stringy gravitons form collective – i.e., ‘coherent’ – states; and ...
- (iii) that a classical metric field corresponds to such coherent quantum states, in the classical limit.

Additionally, when considering the derivation of GR from string theory, Huggett and Vistarini (2015, p. 1169) claim that:

[T]he results do not simply show that strings tell matter and geometry backgrounds how to behave: they ... describe how fields built from string excitations are related, so that the [Einstein] field equations are simply low energy descriptions of the string itself. ... [Ultimately] ... the general relativistic theory of spacetime is a low energy effective strings theory.⁵⁵

As such, string theory is potentially another case of hierarchical emergence, where spacetime emerges from an underlying string-theoretic structure.

Now I will apply (1) the dependence criterion. In light of Huggett and Vistarini’s (2015, p. 1169) claims regarding the derivation of GR from string theory and Huggett and Wüthrich (2025) outline of the approximation of spacetime via coherent states that, in the classical limit, approximate classical spacetime, criterion (1) can plausibly be fulfilled. This is so because if (we provisionally assume) string theory is true and conditionalise on some string theory base details or equivalence class of these (we need not take any interpretive stance on these though), the relevant string theoretic states (e.g. our coherent states), under the right (low-energy) conditions successfully approximate spacetime (where these can, in turn, be usefully modelled via the classical metric of GR). Thus, the spacetime that we recover from string theory at low energies, would not exist or behave the way it does without this underlying string-theoretic structure, and thus depends on this. As such, the relevant string theory (lower-level) base details are unconditionally relevant, fulfilling (1).

Thus, if E_2 represent a classical spatiotemporal description and B the stringy base details, the probability of the emergent general relativistic spacetime description E_2 increases: $P(E_2|B) > P(E_2)$. This is because, at sufficiently low energies, should the conditions Huggett and Wüthrich (2025) describe be fulfilled, the probability of emergent spacetime is plausibly 1. However, if we consider strings at extremely high energies, such a picture approximating classical spacetime is potentially absent, should the conditions Huggett and Wüthrich list not be fulfilled. Ultimately, as “at the quantum string length scale, gravitons are dynamical byproducts of quantum string dynamics ... [yet] ... at ordinary large-distance scales, gravitons’ collective behaviour (their coherent superposition) obeys general relativistic laws” (Vistarini, 2019, p. 56). So, it makes sense that stringy dynamics are unconditionally relevant to the

⁵⁵ For a philosophically oriented overview of this derivation, see Vistarini (2019).

emergence of spacetime, and that taking these into account increases the probability of recovering a classical general relativistic picture.⁵⁶

Criterion (2) (autonomy/conditional irrelevance) may also be met. As Dawid (2007, p. 6) summarises:

The obvious fact that only 4 space-time dimensions are macroscopically visible is taken into account by the assumption that 6 dimensions are ‘compactified’. They have the topological shape of a cylinder surface where, after some translation in the ‘compactified’ direction, one ends up again at the point of departure. The compactification radius as well as the string length are assumed to be so small that both the extension of the string and the additional dimensions are invisible to the current particle experiments. Both scales are expected to lie close to the so-called Planck length.

One might think that compactification and small string length suffice to secure (approximate) conditional irrelevance and fulfil (2). However, the precise nature of the compactification relates to the aforementioned duality issue, as in the case of T-duality, this duality involves the radius of compactified dimensions.⁵⁷ Thus, little specific can be said about this (and string theory’s fundamental ontology) without supplying an interpretation of dualities (issues I remain neutral on). However, string theory minimally involves generic quantum features such as superposition and entanglement. So, at least, generic quantum effects must be screened off and (approximately) conditionally irrelevant at the scales classical GR is well-tested (indeed, they have to be for string theory to be empirically adequate).⁵⁸ To the extent that any higher (than four) dimensional physics features in string theory’s fundamental ontology, this, too, should be conditionally irrelevant at macroscopic scales (with compactification giving a general story about how this screening-off occurs). Again, (like in LQG), the ability to use GR effectively, without regard to stringy lower-level details in regimes where GR is well tested, is strong evidence of the approximate conditional irrelevance of these stringy base details (whatever they are).⁵⁹ Thus, general relativistic spacetime can be understood as autonomous from the (conditionally irrelevant) stringy base details. Consequently, criterion (2), the autonomy condition, appears to

⁵⁶ Additionally, that unconventional, non-gravitational, so-called ‘little string theories’ exist (Aharony, 2000), also indicates that the specific base details are clearly relevant to the probability of a successful recovery, or not, of GR in a broadly string-theoretic context.

⁵⁷ T-duality relates two empirically equivalent theories, one where the radius of the compact dimension is R , the other where it is l^2/R , if l represents the string length (Dawid, 2013).

⁵⁸ An anonymous reviewer rightly points out that such standard quantum features are also screened off in other QG approaches, too. I agree, but make a point of highlighting these here to indicate that even if we do not focus on the notion of compactification as an account of screening higher-dimensional physics in string theory, a minimal form of emergence, where the generic quantum aspects of string theory are conditionally irrelevant at macroscopic scales where classical GR applies, is present in string theory, such that the conditional probability-based plausibly approach applies here.

⁵⁹ So just like in the LQG case, the condition $P(E_2|E_1 \& B) \approx P(E_2|E_1)$ is fulfilled, if B represent our stringy base details and E_1 - E_2 a macrodependency described via GR, such as a precession of the perihelion of mercury calculation, where attending to the stringy base detail is unneeded.

be satisfied in this case, and in a way that remains neutral regarding string theory's underlying ontology.⁶⁰

In string theory, the novelty criterion is trivially satisfied, given the obvious formal distinctness between stringy quantum dynamics, the presence of an interconnected web of dual theories, and ten, or even eleven-dimensional physics (and the corresponding high-energy stringy DOFs, although we can remain neutral regarding precisely what they are), from its classical four-dimensional general relativistic counterpart.

5.4 Flat emergence and Quantum Graphity

We also need to establish that the conditional probability-based approach fits flat emergence.

First, let us consider the dependence criterion (1). Recall Sect. 2.1's quantum graphity example. Here, we can consider the less interconnected and translation symmetric post-geometrogenesis state our emergent state E (more precisely, consider a dependency relation between such states E_1 - E_2). In turn, the base details are the pre-geometrogenesis base state, a maximally-connected permutation invariant state 'prior' to any post-geometrogenesis state E . We have dependence because no change occurs in the emergent state without a change in the pre-geometrogenesis base details, but "not vice versa" (Crowther, 2022, p. 397). This results from the temporal aspect of the theory as the pre-geometrogenesis base state is prior to the E state (or any E_1 - E_2 post-geometrogenesis dependencies of states) "via the theory's evolution equation" and all are states of the one system: the universe (Crowther, 2022, p. 397). Thus, conditional on particular pre-geometrogenesis (base) details, the probability of the state E_2 (of our post-geometrogenesis dependency) increases, as it cannot exist or behave as it does without the prior base details, implying the unconditional relevance of these (fulfilling (1)). Thus, the probability of the emergent description E_2 plausibly increases: $P(E_2|B) > P(E_2)$, as without the base details, we have no emergent dependency/description.

Regarding novelty, that our emergent state E_2 is non-permutation invariant but translation invariant, and the pre-geometrogenesis details are permutation invariant, this symmetry difference provides formal distinctness and, thus, novelty. As for the autonomy criterion (2), that our emergent state (or dependency of these E_1 - E_2) is only dependent "on the broken symmetry pattern that the system instantiates", not on the permutation symmetric pre-geometrogenesis (base) details (Crowther, 2022, p. 398), plausibly implies that these permutation symmetric pre-geometrogenesis base details are approximately conditionally irrelevant post-geometrogenesis.⁶¹ The links that are switched off during the geometrogenesis phase transition are also base details that

⁶⁰To further illustrate this, the precise stringy base details are conditionally irrelevant when we conditionalise on a particular macrostate E_1 (when using GR to describe a macroscopic relationship between spatiotemporal states E_1 - E_2) when applying our autonomy criterion (2). Thus, we require no specific commitment regarding such stringy base (lower-level) details' ontology, only grounds to think that screening-off is present, which is evidenced by the fact that GR works in such regimes.

⁶¹So again, like in the LQG case, the autonomy condition $P(E_2|E_1 \& B) \approx P(E_2|E_1)$ is plausibly fulfilled, if B represent our base details and E_1 - E_2 a post-geometrogenesis macrodependency described via GR.

are plausibly approximately conditionally irrelevant post-geometrogenesis, such that our emergent state/dependency displays autonomy. Thus, the conditional probability-based approach plausibly applies in this case. However, more work is required to establish its viability in other flat emergence cases.

6 Conclusion

In sum: four desiderata for generally applicable accounts of emergence in QG were first established. Wilson's DOF-based emergence approach faces issues due to its commitment to causal metaphysics (which conflicts with (iv), the metaphysical neutrality desideratum), as does Bain's for different reasons, and Bain's view also does not accommodate flat emergence. Crowther (2021, 2022) view is on the right track, but it is comparatively unclear compared to the conditional-probability-based approach, which was the next option introduced. Examples were then given to establish the viability of the conditional-probability-based approach in QG. This fares comparably well regarding our desiderata. Consequently, the conditional probability-based approach is the preferable option if we seek an account of dependence, novelty, and autonomy that fulfils the desiderata and is maximally broadly applicable in QG, although the other accounts under consideration here are still useful in some contexts.

Acknowledgements I would like to thank my PhD supervisor Sam Baron and the various participants in the philosophy of physics reading group that he organised for their feedback on this paper.

Author contribution I am the sole author.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions. No funding to disclose.

Data availability No data to be made available.

Declarations

Disclosures The author has nothing to disclose.

Conflict of interest Not Applicable. In other words, there are no conflicts of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Aharony, O. (2000). A brief review of ‘little string theories’. *Classical and Quantum Gravity*, 17(5), 929–938. <https://doi.org/10.1088/0264-9381/17/5/302>
- Bain, J. (2013a). Emergence in effective field theories. *European Journal for Philosophy of Science*, 3(3), 257–273. <https://doi.org/10.1007/s13194-013-0067-0>
- Bain, J. (2013b). The emergence of spacetime in condensed matter approaches to quantum gravity. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 44(3), 338–345. <https://doi.org/10.1016/j.shpsb.2012.05.001>
- Baron, S. (2020). The curious case of spacetime emergence. *Philosophical Studies: An International Journal for Philosophy in the Analytic Tradition*, 177(8), 2207–2226. <https://doi.org/10.1007/s11098-019-01306-z>
- Baron, S. (2023). Eliminating spacetime. *Erkenntnis*, 88(3), 1289–1308. <https://doi.org/10.1007/s10670-021-00402-z>
- Baron, S., & Le Bihan, B. (2021). Quantum gravity and Mereology: Not so simple. *The Philosophical Quarterly*, 72(1), 19–40. <https://doi.org/10.1093/pq/pqab016>
- Baron, S., & Le Bihan, B. (2024). Causal theories of spacetime. *Noûs*, 58(1), 202–224. <https://doi.org/10.1111/nous.12449>
- Baron, S., & Miller, K. (2015). Causation sans time. *American Philosophical Quarterly*, 52, 27–40.
- Baron, S., Miller, K., & Tallant, J. (2022). *Out of time: A philosophical study of timelessness*. Oxford University Press.
- Baron, S., & Pohlmann, J. (2025). Priority monism and the emergence of spacetime. *Synthese*, 205(2), 1–23. <https://doi.org/10.1007/s11229-024-04859-7>
- Blanchard, T. (2016). Physics and causation. *Philosophy Compass*, 11(5), 256–266. <https://doi.org/10.1111/phc3.12319>
- Butterfield, J. (2011). Less is different: Emergence and reduction reconciled. *Foundations of Physics*, 41(6), 1065–1135. <https://doi.org/10.1007/s10701-010-9516-1>
- Crowther, K. (2015). Decoupling emergence and reduction in physics. *European Journal for Philosophy of Science*, 5(3), 419–445. <https://doi.org/10.1007/s13194-015-0119-8>
- Crowther, K. (2016). *Effective spacetime: Understanding emergence in effective field theory and quantum gravity*. Springer.
- Crowther, K. (2018). Inter-theory relations in quantum gravity: Correspondence, reduction, and emergence. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 63, 74–85. <https://doi.org/10.1016/j.shpsb.2017.12.002>
- Crowther, K. (2021). ‘As below, so before’: ‘synchronic’ and ‘diachronic’ conceptions of spacetime emergence. *Synthese*, 198(8), 7279–7307. <https://doi.org/10.1007/s11229-019-02521-1>
- Crowther, K. (2022). Spacetime emergence: Collapsing the distinction between content and context? In S. Wuppuluri, & I. Stewart (Eds.), *From electrons to elephants and elections: Saga of content and context*. Springer.
- Dawid, R. (2007). Scientific realism in the age of string theory. *Physics and Philosophy*, 11, 3584.
- Dawid, R. (2013). *String theory and the scientific method*. Cambridge University Press.
- Ellis, G. F. R. (2020). Emergence in solid state Physics and biology. *Foundations of Physics*, 50(10), 1098–1139. <https://doi.org/10.1007/s10701-020-00367-z>
- Farr, M., & Reutlinger, A. (2013). A relic of a bygone age? Causation, time symmetry and the directionality argument. *Erkenntnis*, 78(S2), 215–235. <https://doi.org/10.1007/s10670-013-9510-z>
- Franklin, A. (2024). Incoherent? No, just decoherent: How quantum many worlds emerge. *Philosophy of Science*, 91(2), 288–309. <https://doi.org/10.1017/psa.2023.155>
- Franklin, A., & Knox, E. (2018). Emergence without Limits: The case of Phonons. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 64, 68–78. <https://doi.org/10.1016/j.shpsb.2018.06.001>
- Franklin, A., & Robertson, K. (2024). Emerging into the rainforest: Emergence and special science ontology. *European Journal for Philosophy of Science*, 14(4), 1–25. <https://doi.org/10.1007/s13194-024-00622-4>
- Frisch, M. (2023). Causation in physics. *The stanford encyclopedia of philosophy*.
- Grimmer D Cinti, D., Cinti, E., & Jakslund, R. (2024). Duality, underdetermination, and the uncommon common core. *British Journal for the Philosophy of Science*. <https://doi.org/10.1086/730421>

- Guay, A., & Sartenaer, O. (2016). A new look at emergence. Or when after is different. *European Journal for Philosophy of Science*, 6(2), 297–322. <https://doi.org/10.1007/s13194-016-0140-6>
- Huggett, N. (2017). Target space $\neq$ space. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 59, 81–88. <https://doi.org/10.1016/j.shpsb>
- Huggett, N., & Wüthrich, C. (2013). Emergent spacetime and empirical (in) coherence. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 44(3), 276–285. <https://doi.org/10.48550/arXiv.1206.6290>
- Huggett, N., & Wüthrich, C. (2025). *Out of nowhere: The emergence of spacetime in theories of quantum gravity*. Clarendon Press.
- Huggett, N., & Vistarini, T. (2015). Deriving general relativity from string theory. *Philosophy of Science*, 82(5), 1163–1174. <https://doi.org/10.1086/683448>
- Humphreys, P. (2016). *Emergence: A Philosophical account*. Oxford Academic.
- Ismael, J. (2021). Do you see space? How to recover the visible and tangible reality of space (without space). In C. Wüthrich, B. Le Bihan, & N. Huggett (Eds.), *Philosophy beyond spacetime: Implications from quantum gravity*. Oxford University Press.
- Jaksland, R., & Salimkhani, K. (2023). *The many problems of spacetime emergence in quantum gravity*. <https://www.journals.uchicago.edu/>. <https://www.journals.uchicago.edu/>. <https://doi.org/10.1086/727052>
- Knox, E. (2016). Abstraction and its Limits: Finding space for novel explanation. *Noûs*, 50(1), 41–60. <https://doi.org/10.1111/nous.12120>
- Konopka, T., Markopoulou, F., & Severini, S. (2008). Quantum graphity: A model of emergent locality. *Physical Review D*, 77(10), 1–15. <https://doi.org/10.1103/PhysRevD.77.104029>
- Ladyman, J., & Ross, D. (2007). *Every thing must go: Metaphysics naturalized*. Oxford University Press.
- Lam, V., & Esfeld, M. (2013). A dilemma for the emergence of spacetime in canonical quantum gravity. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 44(3), 286–293. <https://doi.org/10.1016/j.shpsb.2012.03.003>
- Lam, V., & Oriti, D. (2024). The quantum gravity seeds for laws of nature. *European Journal for Philosophy of Science*, 14(4), 1–23. <https://doi.org/10.1007/s13194-024-00626-0>
- Lam, V., & Wüthrich, C. (2018). Spacetime is as spacetime does. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 64, 39–51. <https://doi.org/10.1016/j.shpsb.2018.04.003>
- Lam, V., & Wüthrich, C. (2023). Laws beyond spacetime. *Synthese*, 202(3), 1–24. <https://doi.org/10.1007/s11229-023-04305-0>
- Le Bihan, B. (2018). Space emergence in contemporary physics: Why we do not need fundamentality, layers of reality and emergence. *Disputatio*, 10(49), 71–95. <https://doi.org/10.2478/disp-2018-0004>
- Le Bihan, B. (2020). String theory, loop quantum gravity and eternalism. *European Journal for Philosophy of Science*, 10(2). <https://doi.org/10.1007/s13194-020-0275-3>
- Le Bihan, B., & Read, J. (2018). Duality and ontology. *Philosophy Compass*, 13(12). <https://doi.org/10.1111/phc3.12555>
- Legg, C., & Meyer, A. (2026). Modal structure and Sellar's metaphysical methodology', forthcoming. In L. Kocsis, & K. Pete (Eds.), *Wilfrid Sellars's metaphilosophy: Two images and the philosophy in between*. Bloomsbury Academic.
- Livine, E. R. (2024). Spinfoam models for quantum gravity: Overview. In *Encyclopedia of mathematical physics* (2nd ed., Vol. 1, pp. 507–519). <https://doi.org/10.1016/B978-0-323-95703-8.00253-6>. <https://arxiv.org/pdf/2403.09364>
- Matsubara, K. (2013). Realism, underdetermination and string theory dualities. *Synthese*, 190(3), 471–489. <https://doi.org/10.1007/s11229-011-0041-3>
- Ney, A. (2022). Review of metaphysical emergence by Jessica Wilson. *Notre Dame Philosophical Reviews*.
- Norton, J. (2020). Loop quantum ontology: Spacetime and spin-networks. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 71, 14–25. <https://doi.org/10.1016/j.shpsb.2020.04.001>
- Price, H., & Corry, R. (2007). *Causation, Physics, and the Constitution of reality*. Russell's Republic Revisited, Clarendon Press.
- Read, J., & Möller-Nielsen, T. (2020). Motivating dualities. *Synthese*, 197(1), 263–291. <https://doi.org/10.1007/s11229-018-1817-5>
- Rickles, D. (2017). Dual theories: 'same but different' or 'different but same'? *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 59, 62–67. <https://doi.org/10.1016/j.shpsb.2015.09.005>

- Rivat, S., & Grinbaum, A. (2020). Philosophical foundations of effective field theories. *The European Physical Journal A*, 56(3). <https://doi.org/10.1140/epja/s10050-020-00089-w>
- Rovelli, C. (2004). *Quantum gravity*. Cambridge University Press.
- Rovelli, C. (2020). Space and time in loop quantum gravity. In N. Huggett, K. Matsubara, & C. Wüthrich (Eds.), *Beyond Spacetime*, Cambridge University Press.
- Rovelli, C. (2023). How oriented causation is rooted into thermodynamics. *Philosophy of Physics*, 1(1), 1–14. <https://doi.org/10.31389/pop.46>
- Rovelli, C., & Vidotto, F. (2014). *Covariant loop quantum gravity*. Cambridge University Press.
- Russell, B. (1913). On the notion of cause. *Proceedings of the Aristotelian Society*, 13(1), 1–26. <https://doi.org/10.1093/aristotelian/13.1.1>
- Stone, A. (2015). The politics of clarity. *Hypatia*, 30(3), 613–619. <https://doi.org/10.1111/hypa.12143>
- Tang, Z. (2023). Timeless causation? *Acta Analytica*, 38(3), 471–479. <https://doi.org/10.1007/s12136-022-00520-5>
- Thiemann, T. (2007). *Modern canonical quantum general relativity*. Cambridge University Press.
- Vistarini, T. (2019). *The emergence of spacetime in string theory*. Routledge.
- Wallace, D. (2022). Stating structural realism: Mathematics-first approaches to physics and metaphysics. *Nous-Supplement: Philosophical Perspectives*, 36(1), 345–378. <https://doi.org/10.1111/phpe.12172>
- Wallace, D. (2024a). Real patterns in Physics and beyond. <https://philsci-archive.pitt.edu/23307/>
- Wallace, D. (2024b). Learning to represent: Mathematics-first accounts of representation and their relation to natural language. <https://philsci-archive.pitt.edu/23224/>
- Williamson, T. (2006). Must do better. In P. Greenough, & MP. Lynch (Eds.), *Truth and realism*. Oxford University Press.
- Wilson, J. (2010). Non-reductive physicalism and degrees of freedom. *British Journal for the Philosophy of Science*, 61(2), 279–311. <https://doi.org/10.1093/bjps/axp040>
- Wilson, J. (2013). Nonlinearity and metaphysical emergence. In S. Mumford, & M. Tugby (Eds.), *Metaphysics and science*. Oxford Academic. <https://doi.org/10.1093/acprof:oso/9780199674527.003.0010>
- Wilson, J. (2015). Metaphysical emergence: Weak and strong. In T. Bigaj, & C. Wuthrich (Eds.), *Metaphysics in contemporary physics*. Brill.
- Wilson, J. (2020). *Metaphysical emergence*. Oxford University Press.
- Wilson, J. (2024). On the notion of diachronic emergence. In A. Bryant, & D. Yates (Eds.), *Rethinking emergence*. Oxford University Press.
- Woodward, J. (2007). Causation with a human face. In H. Price, & R. Corry (Eds.), *Causation, physics, and the constitution of reality. Russell's republic Revisited*. Clarendon Press.
- Wüthrich, C. (2017). Raiders of the lost spacetime. In D. Lehmkuhl, G. Schieman, & E. Scholz *Towards a theory of spacetime theories*. Springer. https://doi.org/10.1007/978-1-4939-3210-8_11
- Zwiebach, B. (2009). *A first course in string theory*. Cambridge University Press.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.